

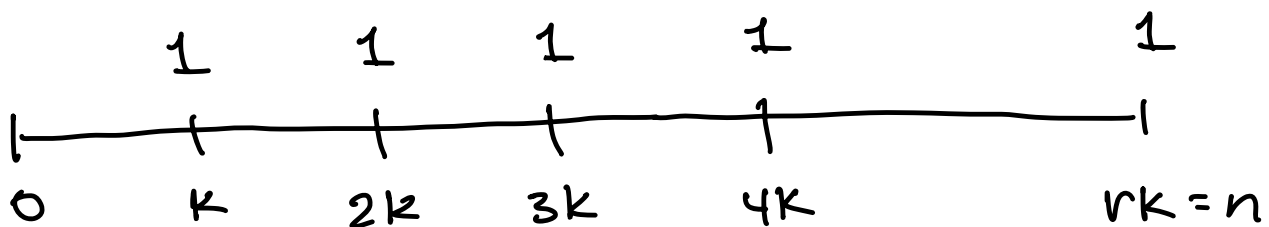
4.2 Level Annuities with payments less frequent than each interest period

Suppose we have an annuity w/ annual pmts but a monthly interest rate is given

option 1 $\rightarrow$ Convert the rate!!!

Now we learn option 2 which is

- (a) More computationally efficient
- (b) aesthetically appealing



(Think $K=12$ if monthly rate is given)

If " i " is a monthly rate then the rate we want say " I " is

$$I = (1+i)^K - 1 \quad \text{and we want } a_{\overline{r}I}$$

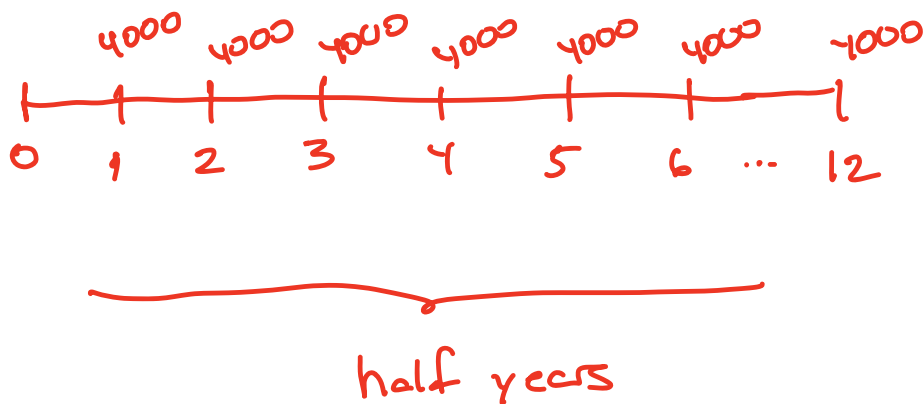
$$a_{\overline{n}|i} = \frac{1 - (1+i)^{-n}}{i} = \frac{1 - (1 + ((1+i)^k - 1))^{-n}}{(1+i)^k - 1}$$

$$= \frac{1 - (1+i)^{-kn}}{(1+i)^k - 1} = \frac{1 - (1+i)^{-n}}{(1+i)^k - 1}$$

$$= \frac{1 - (1+i)^{-n}}{i} \cdot \frac{i}{(1+i)^k - 1} = \frac{a_{\overline{n}|i}}{s_{\overline{k}|i}}$$

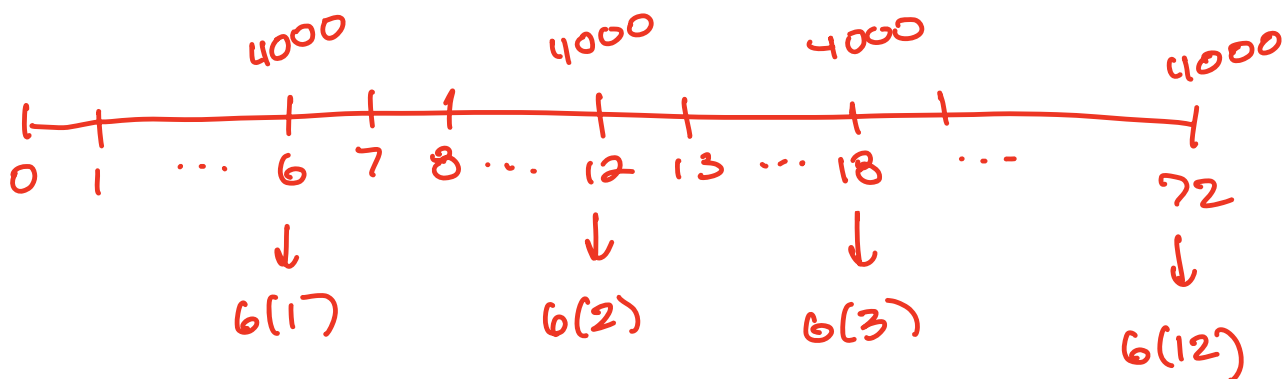
Problem: Chinh is presented an annuity that is to pay \$4,000 at the end of each six months for six years. Using compound interest at a nominal monthly interest rate of 3%, what is the value of Chinh's reward at the time of the announcement (six months before the first \$4,000 payment)?

solution.



but $i^{(12)} = 3\% \Rightarrow \frac{i^{(12)}}{12} = .0025$

In months: $k = 6, n = 72$

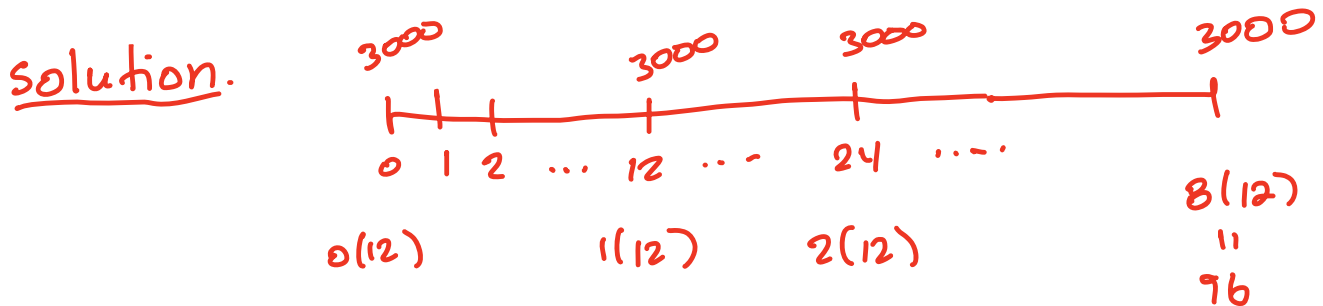


$$PV = \frac{4000 \cdot a_{\overline{72}|.0025}}{s_{\overline{6}|.0025}} = 43,604.467111$$

If we consider annuity due:

$$\ddot{a}_{\overline{n}|I} = (1+I) a_{\overline{n}|I} = (1+i)^k \frac{a_{\overline{n}|i}}{S_{\overline{k}|i}} = \frac{a_{\overline{n}|i}}{a_{\overline{k}|i}}$$

Problem: Josh makes annual beginning-of-year contributions of \$3,000 to a retirement account. The account has a nominal interest rate of 4.8% convertible monthly. Find Josh's balance just before his ninth contribution by finding the annual effective interest rate and then check your answer using (4.2.8).



$$i^{(12)} = 4.8\%$$

$$\text{want } \ddot{S}_{\overline{8}|I} (3000)$$

$$= \frac{a_{\overline{96}|.004}}{a_{\overline{12}|.004}} (1.004)^{96} \cdot 3000 = 29953.297$$

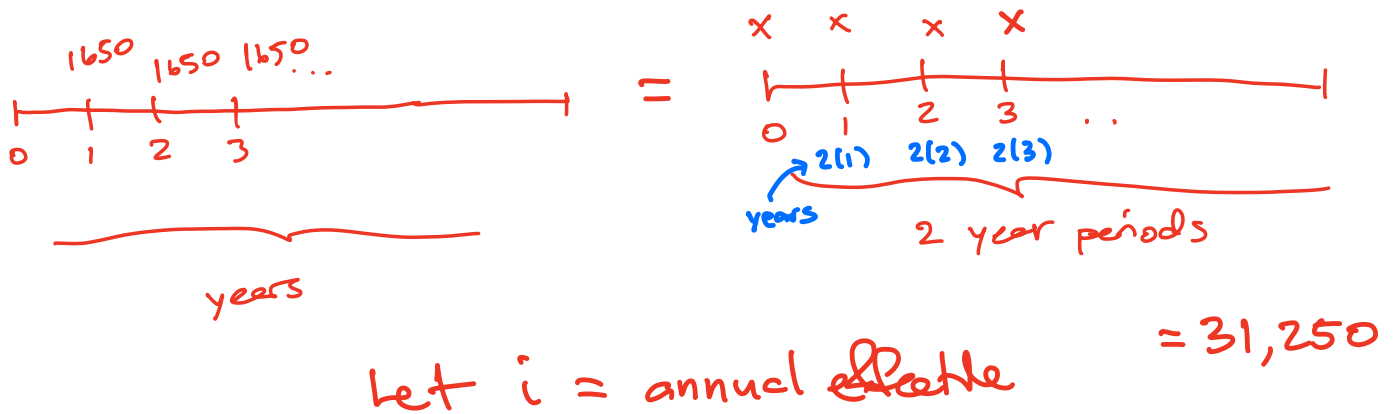
If we think of a perpetuity we have

$$a_{\infty|I} = \frac{1}{I} = \frac{1}{(1+i)^k - 1} = \frac{1}{i s_{\overline{k}|i}}$$

$$\ddot{a}_{\infty|I} = \frac{1}{d} = \frac{1}{\frac{I}{1+I}} = \frac{(1+i)^k}{(1+i)^k - 1} = \frac{1}{i a_{\overline{k}|i}}$$

Problem: A perpetuity paying \$1,650 at the end of each year is to be replaced by another paying X at the beginning of each two years. The present value of each perpetuity is \$31,250. Find X .

Solution.



$$\frac{1650}{i} = 31,250 \Rightarrow i = .0528$$

$$X \left(\frac{1}{i a_{\overline{2}|i}} \right) = 31,250 \Rightarrow X = 3055.897$$

4.3 Level Annuities with payments more frequent than each interest period

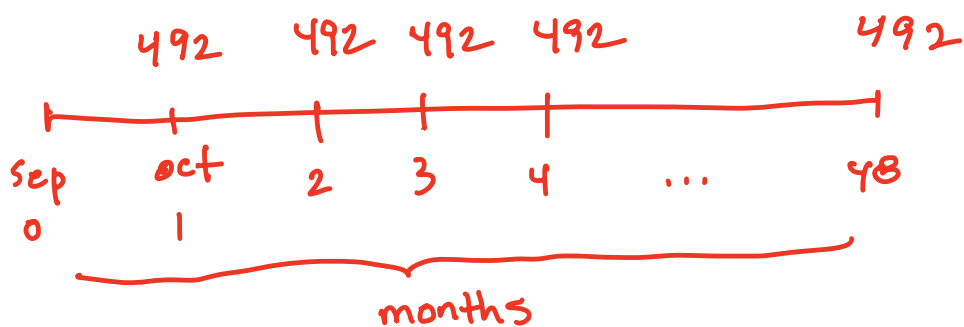
Last section we considered examples where the interest rate given was for a time interval less than, the pmts. Here we consider the opposite.

Problem: Mahpee was offered a scholarship with level monthly payments for four years. The first payment would be made on September 30, 2000, and each year the payments would total \$5,904. Mahpee, who was contemplating actuarial studies, wished to compare this scholarship to a fellowship offered by another college. The latter would pay \$5,800 on September 1st of years 2000, 2001, 2002, and 2003. Assume that the growth of money is governed by compound interest at an annual effective interest rate of 4%, and compare the values of the two grants on September 1, 2000.

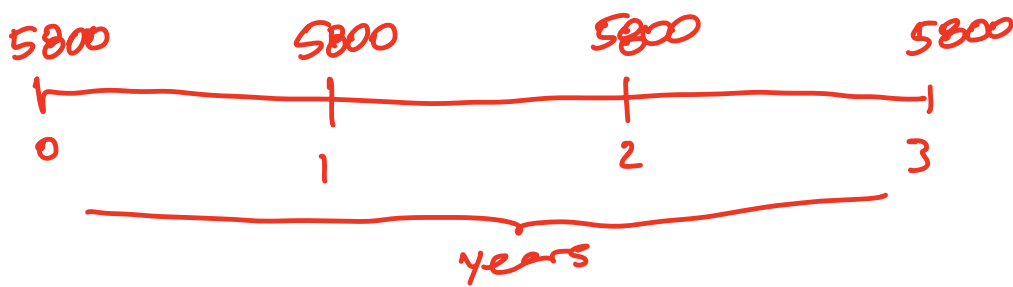
Solution. monthly pmt 4 yrs $\rightarrow$ 48 pmts

$$i = 4\%$$

option 1



option 2



we could just convert $\frac{i^{(12)}}{12} = (1.04)^{1/12} - 1$

$$PV(\text{opt 1}) = 492 a_{\overline{48}|(1.04)^{1/12} - 1} = 21,821.02$$

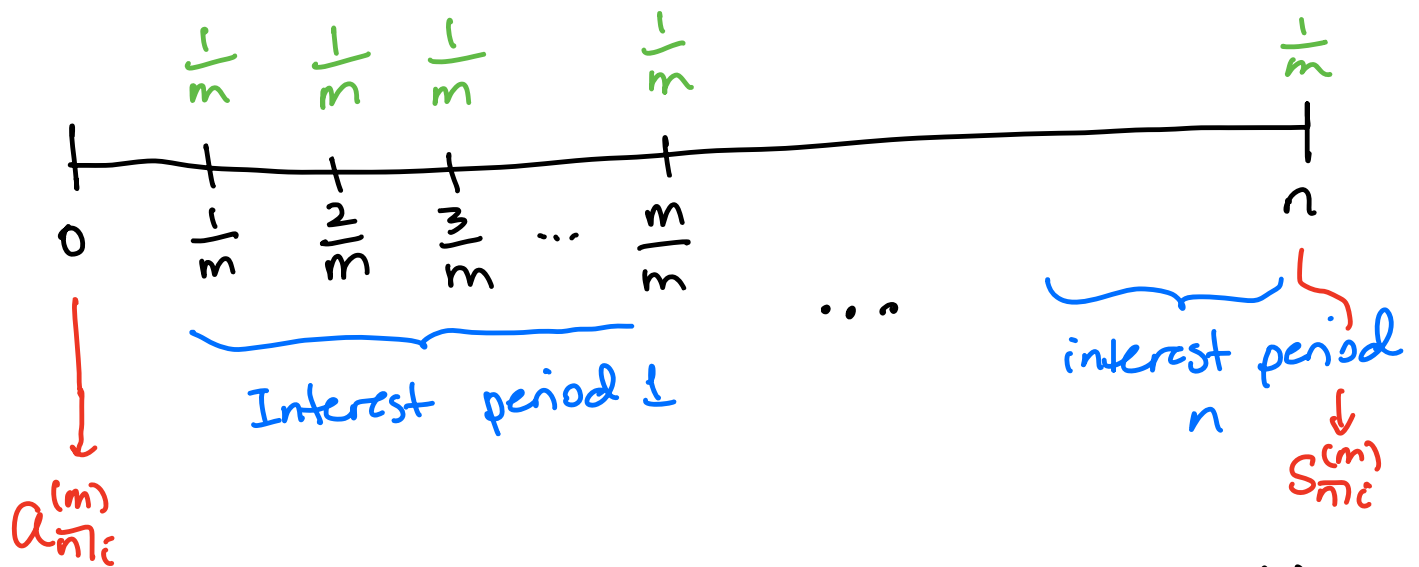
$$PV(\text{opt 2}) = 5800 \left(1 + (1.04)^{-1} + 1.04^{-2} + 1.04^{-3} \right) = 21,895.53$$

opt 2 > opt 1.

We could have written option 1 as

$$5904 \left(\frac{1}{m} \right) a_{\overline{nm}|J} \quad \text{where} \quad \begin{array}{l} m = 12 \\ n = 4 \text{ yrs} \\ J = \text{effective monthly} \\ (1+i)^{1/m} - 1 \end{array}$$

we now analyze $\frac{1}{m} a_{\overline{nm}|J}$ and also
the accumulated value $\frac{1}{m} s_{\overline{nm}|J}$



we use the notation $a_{n|i}^{(m)}$ and $S_{n|i}^{(m)}$ to denote these present and accumulated values at times 0 and n resp.

Easy fact: $a_{n|i}^{(1)} = a_{n|i}$ and $S_{n|i}^{(1)} = S_{n|i}$

can we find closed forms of $a_{n|i}^{(m)}$ in terms of m, n, i ?

Value at $t=0$

$$= \frac{1}{m} V^{\frac{1}{m}} + \frac{1}{m} V^{\frac{2}{m}} + \frac{1}{m} V^{\frac{3}{m}} + \dots + \frac{1}{m} V^{\frac{nm}{m}}$$

let $x = V^{1/m}$

$$= \frac{1}{m} \cdot V^{\frac{1}{m}} \left(1 + x + x^2 + \dots + x^{nm-1} \right)$$

$$= \frac{1}{m} \cdot V^{\frac{1}{m}} \left(\frac{1 - x^{nm}}{1 - x} \right)$$

$$= \frac{1}{m} \cdot V^{\frac{1}{m}} \left(\frac{1 - V^n}{1 - V^{1/m}} \right)$$

$$1 - d = \left(1 - \frac{d^{(m)} m}{m} \right)$$
$$(1 - d)^{1/m} = 1 - \frac{d^{(m)}}{m}$$

what is $\frac{V^{1/m}}{m(1 - V^{1/m})}$

remember $V^{1/m} = (1 + i)^{-1/m}$

or

$$V^{1/m} = (1 - d)^{\frac{1}{m}}$$
$$= 1 - \frac{d^{(m)}}{m}$$

$$= \frac{1 - \frac{d^{(m)}}{m}}{m \left(\frac{d^{(m)}}{m} \right)}$$

$$= \frac{1}{i^{(m)}} !!!$$

So...

$$a_{\overline{n}|i}^{(m)} = \frac{1 - v^n}{i^{(m)}} = a_{\overline{n}|i} \cdot \left(\frac{i}{i^{(m)}} \right)$$

similar computation shows

$$S_{\overline{n}|i}^{(m)} = S_{\overline{n}|i} \cdot \left(\frac{i}{i^{(m)}} \right)$$

Problem: Shirley invests \$200 at the end of each quarter for eight years in an account having an annual effective interest rate of 4%. Find Shirley's accumulated value at the time of her last deposit.

Solution. $pmt = 200$ $n = 8$, $m = 4$, $i = 4\%$.

$$\begin{aligned} 800 S_{\overline{8}|.04}^{(4)} &= S_{\overline{8}|.04} \left(\frac{.04}{(1.04)^{1/4} - 1} \right) \cdot 800 \\ &= \left(\frac{1 - 1.04^{-8}}{.04} \right) (1.04)^8 \left(\frac{.04}{(1.04)^{1/4} - 1} \right) \cdot 800 \\ &= 7481.05 \end{aligned}$$

In a very similar fashion we get

$$\ddot{a}_{\overline{n}|i}^{(m)} = a_{\overline{n}|i} \left(\frac{i}{d^{(m)}} \right) \quad \text{and}$$

$$\ddot{s}_{\overline{n}|i}^{(m)} = s_{\overline{n}|i} \left(\frac{i}{d^{(m)}} \right)$$

we get the following inequalities!

$$\ddot{a}_{\overline{n}|i} > \ddot{a}_{\overline{n}|i}^{(m)} > a_{\overline{n}|i}^{(m)} > a_{\overline{n}|i}$$

Problem: Find the numerical values of the four annuity symbols in inequality (4.3.14) when $n = 48$, $m = 12$, and $i = 5\%$.

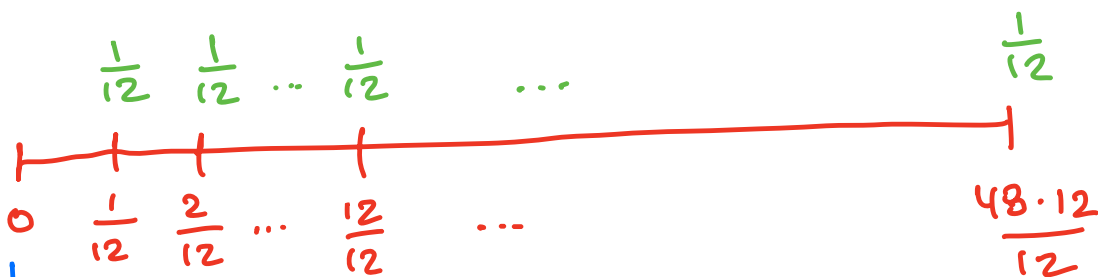
Solution.



$$\ddot{a}_{\overline{n}|i} = \frac{1 - 1.05^{-48}}{.05} (1.05) = 18.981015$$



$$\ddot{a}_{\overline{48}|.05}^{(12)} = \frac{1 - 1.05^{-48}}{.05} \left(\frac{.05}{d^{(12)}} \right) = 18.563098$$



$$a_{\overline{48}|.05}^{(12)} = \frac{1 - 1.05^{-48}}{.05} \left(\frac{.05}{i^{(12)}} \right) = 18.48779674$$



$$\ddot{a}_{\overline{n}|i} = \frac{1 - 1.05^{-48}}{.05} = 18,077.15782$$

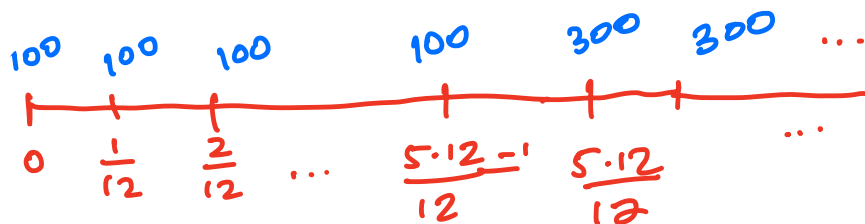
we also have perpetuities

$$a_{\overline{\infty}|i}^{(m)} = \lim_{n \rightarrow \infty} a_{\overline{n}|i}^{(m)} = a_{\overline{\infty}|i} \left(\frac{i}{i^{(m)}} \right) = \frac{1}{i^{(m)}}$$

$$\ddot{a}_{\overline{\infty}|i}^{(m)} = \lim_{n \rightarrow \infty} \ddot{a}_{\overline{n}|i}^{(m)} = \ddot{a}_{\overline{\infty}|i} \left(\frac{i}{d^{(m)}} \right) = \frac{1}{d^{(m)}}$$

Problem: Pavol purchases a perpetuity-due. The perpetuity has monthly payments of \$100 during the first five years, and then these increase to \$300. Determine the purchase price if it is based on an annual effective discount rate of 3.6%.

solution.



$$d = 3.6\%$$

$$\Rightarrow \frac{d^{(12)}}{12} = 1 - .964^{1/12}$$

$$PV = 1200 \ddot{a}_{\overline{5}| \frac{.036}{.964}}^{(12)} + \frac{3600}{.964^{1/12}} (.964)^5$$

$$= 87,358.00$$

4.6 Continuously paying annuities

what is the limiting behavior of $a_{\overline{n}|i}^{(m)}$
which models "continuous pmts"

$$\lim_{m \rightarrow \infty} a_{\overline{n}|i}^{(m)} = \overline{a}_{\overline{n}|i} \quad \leftarrow \text{New notation}$$

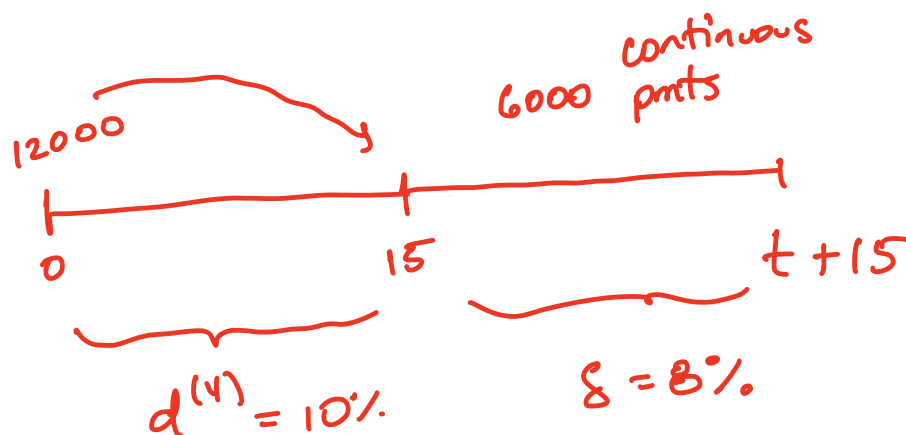
$$\lim_{m \rightarrow \infty} \left(\frac{1 - v^n}{i^{(m)}} \right) = \frac{1 - v^n}{\delta} = \frac{1 - e^{-\delta n}}{\delta}$$

similarly we have

$$\overline{s}_{\overline{n}|i} = \lim_{m \rightarrow \infty} s_{\overline{n}|i}^{(m)} = \frac{e^{\delta n} - 1}{\delta}$$

Problem: Martin deposits \$12,000 in a savings account with a discount rate of 10% convertible quarterly. He leaves his money in this account to accumulate for fifteen years, then moves it to a fund that is accumulating at 8% per annum convertible continuously. If, starting at time 15 when he invests in the new fund, money is withdrawn continuously at a rate of \$6,000 per annum, how long will Martin's money last?

Solution.



$$12000 \left(1 - \frac{.10}{4}\right)^{-60} = \left(\bar{a}_{\overline{t}|i}\right)(6000)$$

$$= \left(\frac{1 - e^{-.08t}}{.08}\right)(6000) \Rightarrow t = 16.41$$

so in year 31.41

$$\ddot{a}_{\overline{n}|i} > \ddot{a}_{\overline{n}|i}^{(m)} > \bar{a}_{\overline{n}|i} > a_{\overline{n}|i}^{(m)} > a_{\overline{n}|i}$$

If we are not working with compound interest, remember

$$a_{\overline{n}|} = \sum_{t=1}^n v(t) \quad \text{so naturally,}$$

$$a_{\overline{n}|}^{(m)} = \sum_{t=\frac{1}{m}}^{nm} v(t) \quad \text{and hence}$$

$$\lim_{m \rightarrow \infty} a_{\overline{n}|}^{(m)} = \overline{a}_{\overline{n}|} = \int_{t=0}^n v(t) dt$$

If $v(t) = v^t$ or $(1+i)^{-t}$ you get

$$\begin{aligned} \overline{a}_{\overline{n}|} &= \int_0^n v^t dt = \frac{v^t}{\ln(v)} \Big|_0^n = \frac{1}{\ln(v)} [v^n - 1] \\ &= \frac{1 - v^n}{\delta} \end{aligned}$$

Fact: If $f(t)$ is a continuous function on $[0, n]$. The P.V. of the cont. paying annuity that pays from time 0 to n at a rate of $f(t)$ at time t is $\int_0^n f(t) v(t) dt$.

Problem: Suppose that money grows according to simple discount at a rate of 3.6%. Find formulas for $\bar{a}_{\overline{n}|}$, $\bar{s}_{\overline{n}|}$, $\ddot{s}_{\overline{n}|}$.

solution. $a(t) = \frac{1}{1 - .036t}$ so $v(t) = 1 - .036t$

$$\begin{aligned}\bar{a}_{\overline{n}|} &= \int_0^n (1 - .036t) dt \\ &= t - .018t^2 \Big|_0^n = n - .018n^2\end{aligned}$$

$$\begin{aligned}\bar{s}_{\overline{n}|} &= \bar{a}_{\overline{n}|} \cdot a(n) \\ &= \frac{n - .018n^2}{1 - .036n}\end{aligned}$$

Problem: An annuity is payable continuously for ten years. The rate of payment at time t is t^2 yearly. Assuming there is a constant force of interest $\delta = .04$, find the present value of the annuity.

Solution. If $\delta_t = .04$ then $a(t) = e^{\int_0^t .04 dr}$
 $= e^{.04t}$

$f(t) = t^2$ so

$$\overline{a}_{10|} = \int_0^{10} t^2 e^{-.04t} dt$$

$u = t^2$ $v = \frac{1}{.04} e^{-.04t}$
 $du = 2t dt$ $dv = e^{-.04t}$

$$\frac{-t^2}{.04} e^{-.04t} + \frac{2}{.04} \int e^{-.04t} t dt$$

$\underbrace{\hspace{10em}}$
 $u = t$ $v = \frac{1}{.04} e^{-.04t}$
 $du = dt$ $dv = e^{-.04t}$

$$\frac{-t}{.04} e^{-.04t} + \frac{1}{.04} \int e^{-.04t} dt$$

$\underbrace{\hspace{10em}}$
 $= \frac{-1}{(.04)^2} e^{-.04t}$

$$= \frac{-t^2}{.04} e^{-.04t} + \frac{2}{.04} \left(\frac{-t}{.04} e^{-.04t} - \frac{1}{.04^2} e^{-.04t} \right) \Bigg|_{n=0}^{n=10}$$

$$= 247.70$$